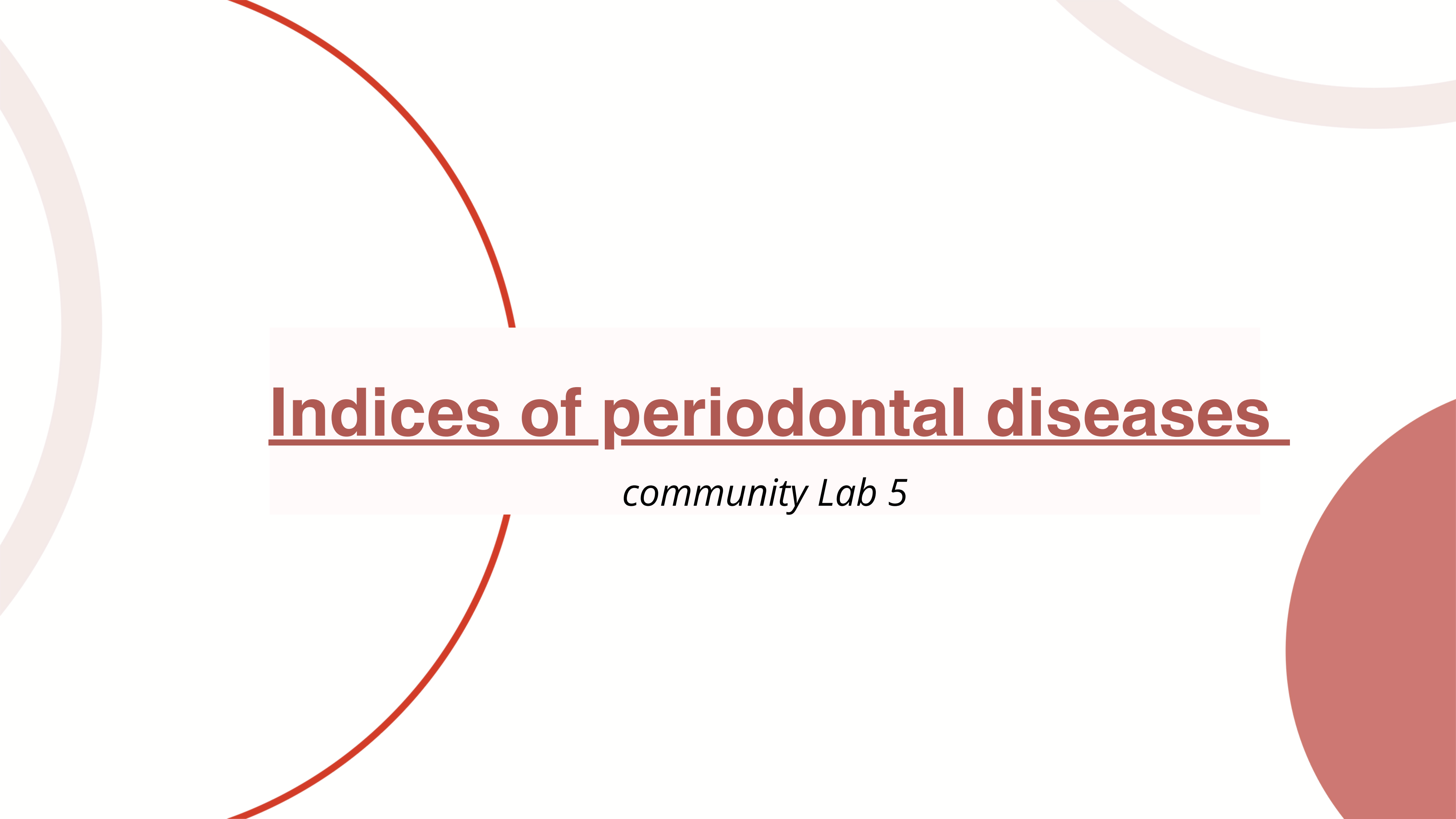




Indices of periodontal diseases

community Lab 5

Healthy periodontium

The periodontium consist of:

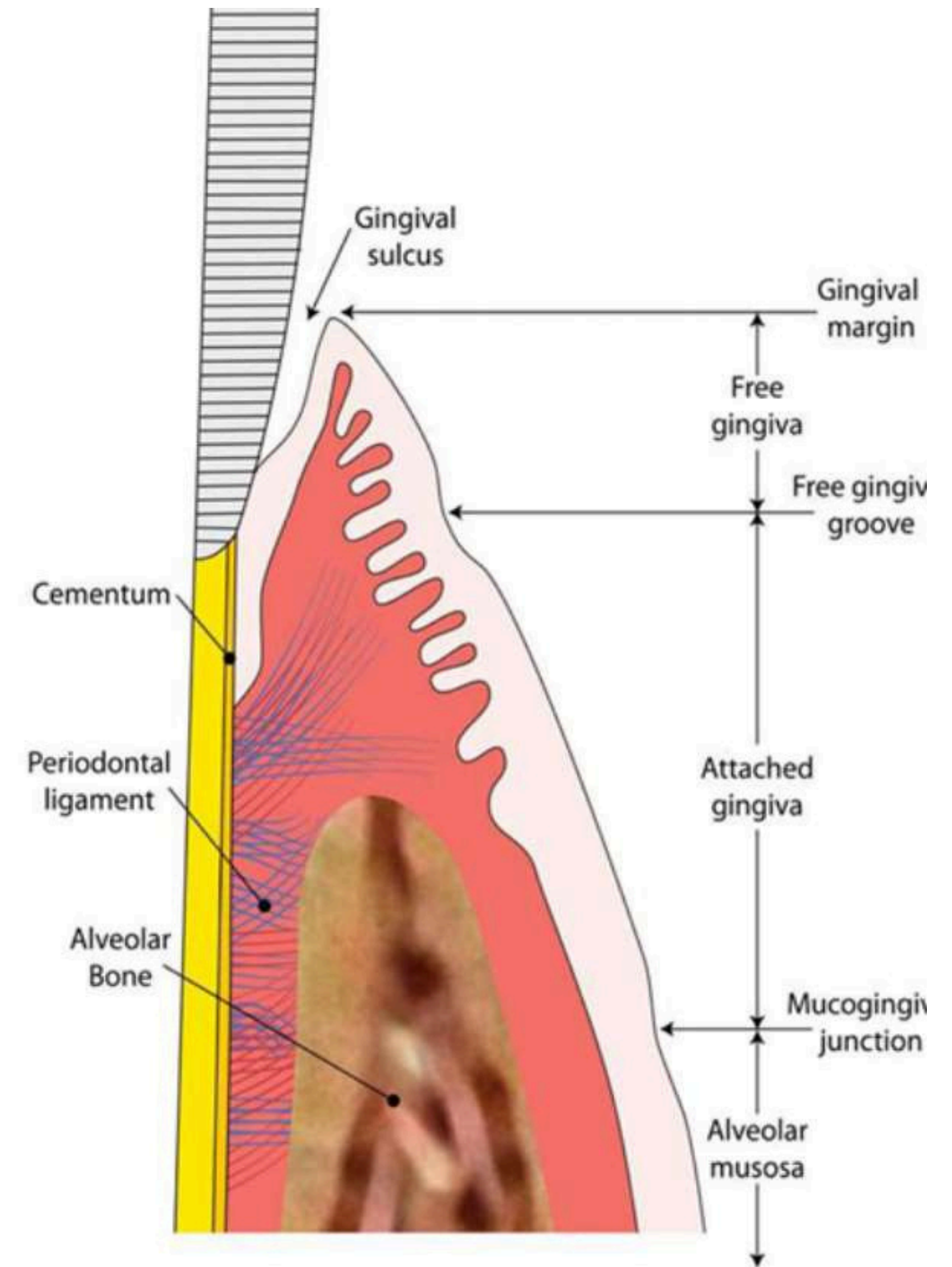
- Gingiva.
- Periodontal ligament.
- Root cementum.
- Alveolar bone (alveolar process).

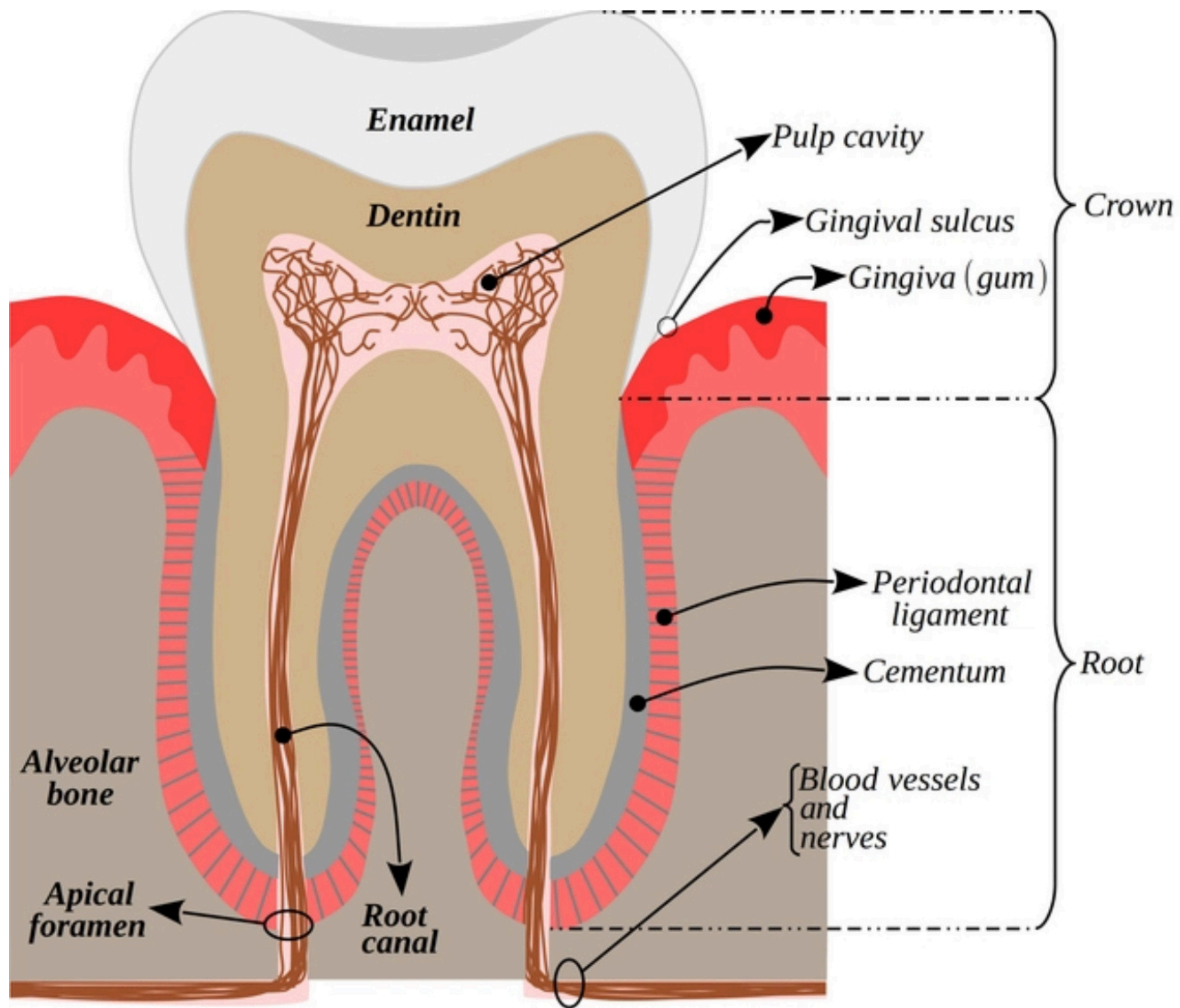
Gingiva

Tissue covering the cervical portions of the teeth and the alveolar processes of the jaws.

The gingiva is demarcated clinically into:

- The free gingival margin (1.5 mm wide).
- The attached gingiva which may be of varying width.
- The interdental gingiva, occupying the place between adjacent teeth.





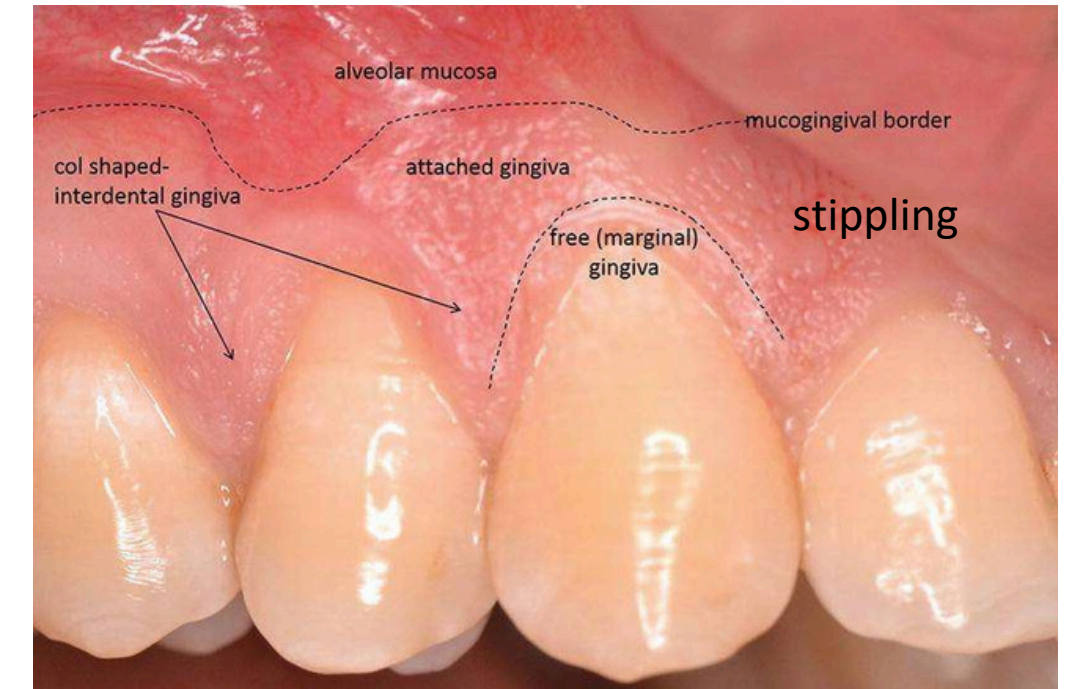
Gingival Sulcus:-

It is the V-shaped space between the free gingiva and the tooth surface.

-The depth is 1-3 mm for a clinically normal gingival sulcus.

Clinical features of healthy gingiva:

- The gingiva is coral pink in color, firm consistency, a scalloped outline and knife edged.
- Often shows small depressions on the surface named "stippling".
- It is firmly attached to the underlying alveolar bone and cementum.



healthy gingiva



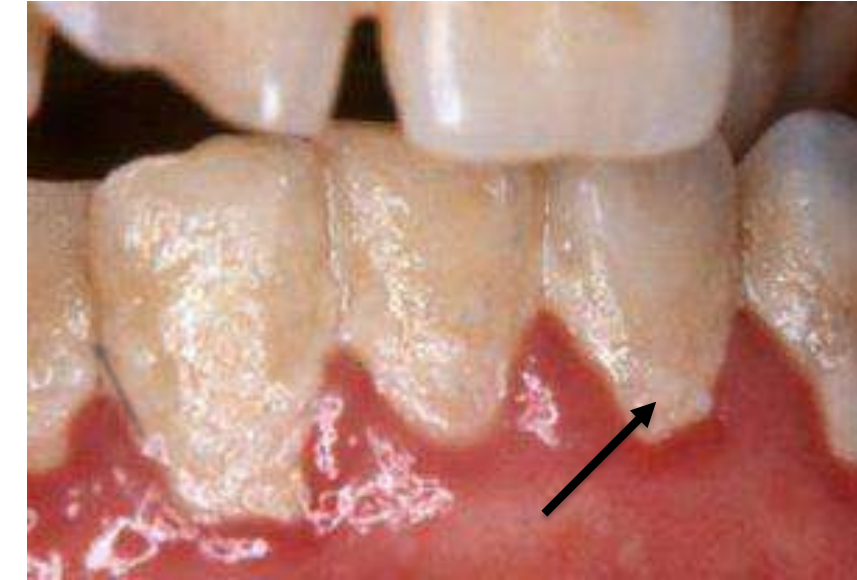
gingivitis

Oral Hygiene Indices:

- Dental plaque is defined as the non-mineralized soft adherent microbial film that adheres tenaciously to the teeth and other solid structures in the mouth.
- Its composed of bacteria in a matrix of salivary glycoproteins and extracellular polysaccharides.
- It resists removal by salivary flow or a gentle spray of water across its surface.

Plaque index (PI I) of Silness and Loe (1964) :

- Each tooth is dried and examined visually using a mirror, an explorer, and adequate light.
- The explorer is passed over the cervical third to check for the presence of plaque (assesses the thickness of plaque at the gingival margin of the tooth).
- Four surfaces (distal, buccal, mesial, and lingual) are examined.
- Each surface of each tooth is given a score from 0 to 3.
- It should be done before examination of GI.
- Missing teeth are not substituted.



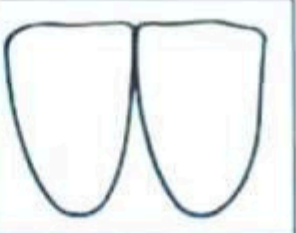
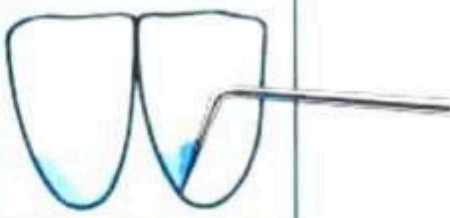
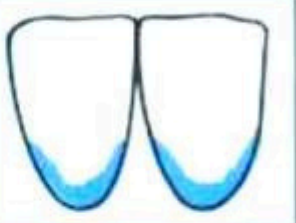
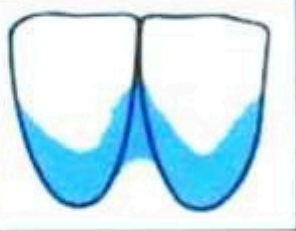
Dental plaque clinical



-Scoring is done on all teeth (28) or selected teeth which are:

6	2	4
4	2	6

Dental plaque is the main pathological cause of periodontal diseases

Grade 0	No plaque	
1	Thin film of plaque at the gingival margin, visible only when scraped with an explorer	
2	Moderate amount of plaque along the gingival margin; interdental space free of plaque; plaque visible with the naked eye	
3	Heavy plaque accumulation at the gingival margin; interdental space filled with plaque	

PI

Calculation of scores:

A – To determine PI for an individual: $\frac{\text{Total scores}}{\text{number of surfaces examined}}$

B- To determine PI for population: $\frac{\text{Total scores for subjects}}{\text{number of subjects examined}}$

C- Suggested nominal scale for patient may be assigned:

- 0 = excellent oral hygiene.
- 0.1-0.9 = good oral hygiene.
- 1.0-1.9 = fair oral hygiene.
- 2.0-3.0 = poor oral hygiene.

Preventive and Community Dentistry / Case sheet

Patient's name: Age:

Student's name: Group:

Oral hygiene and gingival health status

Index	Surface	E	B	D	E	B	D	Mean
		6	2	4	6	2	4	
PII	Buccal							
	Mesial							
	Lingual							
	Distal							
GI	Buccal							
	Mesial							
	Lingual							
	Distal							
Index	Surface	E	A	D	E	A	D	Mean
		6	1	4	6	1	4	
Call	Buccal							
	Lingual							

Plaque Index (Silness and Loe , 1964):

- 0 No plaque.
- 1 Plaque on the gingival 1/3 that cannot be seen by naked eye, only by using probe.
- 2 Moderate accumulation of plaque on the gingival 1/3 that can be seen by naked eye.
- 3 Abundance of plaque on the gingival 1/3.

Gingival Index (Loe and Silness, 1963):

- 0 No inflammation.
- 1 Mild inflammation, slight redness, edema, no bleeding on probing.
- 2 Moderate inflammation, bleeding on probing.
- 3 Severe inflammation, spontaneous bleeding.

Calculus component of Periodontal Disease Index (Ramfjord , 1959):

- 0 No calculus.
- 1 Supra gingival calculus .
- 2 Moderate supra gingival and sub gingival calculus or sub gingival calculus only.
- 3 Abundance of supra gingival and sub gingival calculus .

Calculus component of PDI index of Ramfjord (1959):

-Measure the presence and extension of calculus on the facial and lingual surfaces of the selected teeth.

-Examination is done on selected teeth of Ramfjord which are:

6	1	4
4	1	6

E	A	D
D	A	E

-Score from 0-3.

Scoring criteria:

- 0 Absence of calculus
- 1 Supragingival calculus extending only slightly below the free gingival margin (not more than 1 mm).
- 2 Moderate amount of supragingival and sub gingival calculus or sub gingival calculus alone.
- 3 An abundance of supra gingival and sub gingival calculus.

Calculation:

1- Individual:

$$\text{Cal} = \frac{\text{Total scores}}{\text{No. of surfaces examined}}$$

2- Population:

$$\text{Cal} = \frac{\text{Total scores}}{\text{No. of subjects examined}}$$

Preventive and Community Dentistry / Case sheet

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Oral hygiene and gingival health status

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	Mesial							
	Lingual							
	Distal							
GI	Buccal							
	Mesial							
	Lingual							
	Distal							
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	Lingual							

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Types of calculus:

A-Supra gingival calculus.

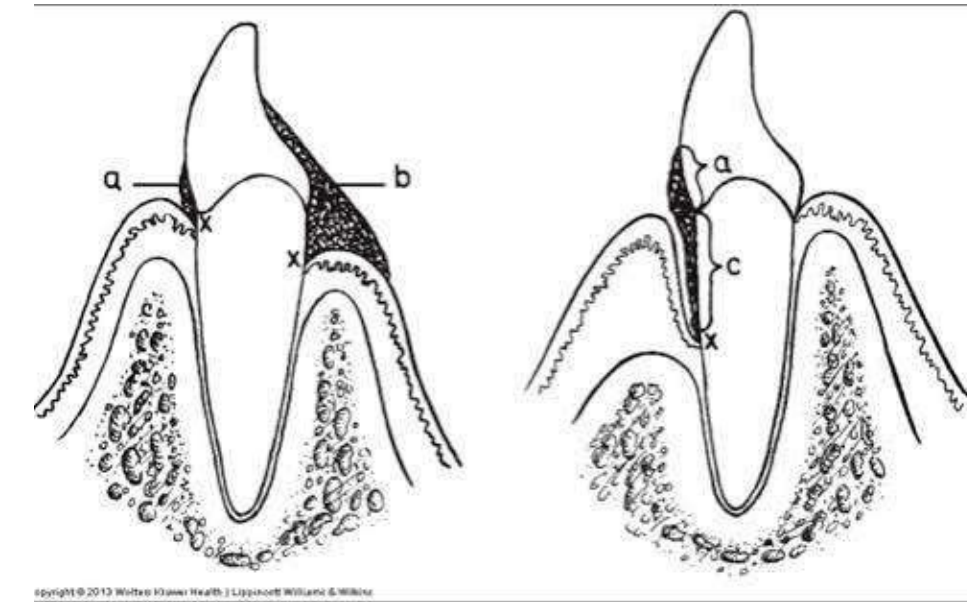
B-Sub gingival calculus.

Differences between sub. and supra gingival calculus:

1_ Sub- gingival calculus is about 60% mineralized while supra gingival calculus is about 30% mineralized

2_ Sub-gingival calculus can be more difficult to remove than supra gingival calculus because of its greater hardness, being thinner and more closely interlinked to tooth surface imperfections.

3_ The supra-gingival calculus usually appears as a yellow - to- white in color whereas the sub gingival calculus appears gray to black. The dark coloration may be due to bacterial degradation of components of hemorrhagic exudates resulting from gingival inflammation.



Gingival Index (GI) of Loe and Silness (1963):

- 1-This index measures the severity of gingivitis based on color, consistency, and bleeding on probing.
- 2- Each tooth is examined at the mesial, lingual, distal and buccal surface.
- 3- Teeth and gingiva are dried and are examined under adequate light, using a mouth mirror and blunt probe.
- 4- A blunt probe is used to press on the gingiva to determine its degree of firmness, and to run along the soft tissue wall adjacent to the entrance to the gingival sulcus to evaluate bleeding.
- 5-The excluded teeth are partially erupted, teeth with periapical lesion and retained roots.
- 5-The index teeth are:

6	2	4
4	2	6

E	B	D
D	B	E

Preventive and Community Dentistry / Case sheet

Patient's name: Age:

Student's name: Group:

Oral hygiene and gingival health status

Index	Surface	E	B	D	E	B	D	Mean
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- 3 Abundance of supra gingival and sub gingival calculus .

Score Criteria

- 0 No inflammation.
- 1 Mild inflammation, slight change in color, slight edema, no bleeding on probing.
- 2 Moderate inflammation, moderate glazing, redness, edema and hypertrophy, bleeding on probing.
- 3 Severe inflammation, marked redness and hypertrophy, ulceration tendency to spontaneous bleeding.



The interpretation:

0.1 - 1 Mild gingivitis

1.1 - 2 Moderate gingivitis

2.1 - 3 Severe gingivitis

Calculation:

1-Individual

$$GI = \frac{\text{Total scores}}{\text{No. of surfaces exa.}}$$

2-Population

$$GI = \frac{\text{Total scores}}{\text{No. of subjects exa.}}$$

-The maximum score for gingival index is :

$$GI = \frac{\text{Total scores}}{\text{No. of surfaces exa.}}$$

$$GI = \frac{3 \times 4 \times 6}{4 \times 6} = \frac{3 \times 24}{24} = 3$$



Thank you